

Mathias Rodríguez Castro
Facultad de Ingeniería, Universidad de la República
Montevideo, Uruguay
mathiasr@fing.edu.uy

To the Editors of *Computational Optimization and Applications*

Dear Editors,

Please consider “Pre-Export Row Scaling in Generated Mixed-Integer Programs: Mechanism Attribution and Residual-Budget Contracts” for publication in *Computational Optimization and Applications*.

Positive row scaling preserves a mixed-integer program in exact arithmetic, but changes its finite-precision behavior and the original-unit meaning of feasibility tolerances. Beyond asking which scaling improves an optimization pipeline, this paper asks which mechanism produces the apparent improvement and which accuracy guarantee survives the change of representation.

Two analytical contributions organize the paper. First, geometric-mean centering eliminates an extrema-based block statistic, making an apparently metadata-dependent coupling rule identical to Euclidean normalization. The locally whitened triangular model admits the closed-form optimal common coupling factor $(1 + \rho^2)^{-1/2}$. Second, residual-budget contracts characterize exactly the factors compatible with application-unit budgets and coefficient limits, including an incompatibility certificate, constrained logarithmic minimax centering, integer-rounding restrictions and conditions for exact binary64 export. Kernel-matched controls distinguish the scaling mechanism from ownership metadata.

Computational validation includes a fixed-basis audit of 306 generated dispatch models, 1,080 matched matrix exports, 240 reconstructed operational runs with two matched sensitivity reruns, and an exactly enumerated stress experiment with 4,752 optimizer calls across three solvers. File-level conditioning gains largely shrink after solver-style equilibration. A completed-run work reduction is reversed by timeout penalties and does not reproduce with a second solver. Adding a residual budget to the same centering rule loses no verified stress optima relative to that rule. These findings support numerical methodology and error analysis; the operational evidence remains limited to one application.

The work connects the journal’s areas of Integer Programming; Approximations and Error Analysis; Software, Benchmarks, Numerical Experimentation and Comparisons; and Modelling Languages and Systems for Optimization. Online Resource 1 provides proofs, protocols and reproduction instructions; Online Resource 2 provides code, stored data, manifests and verification tools.

I am the sole author, received no external funding and declare no competing interests. The manuscript documents AI assistance and my responsibility for its scientific content. The work has not been published and is not under consideration elsewhere.

Sincerely,

Mathias Rodríguez Castro